- Install PostgreSQL

- Connect to the Database

The most common way to interact with PostgreSQL is through its command-line interface, called psql.

•**Windows:** Open the "SQL Shell (psql)" application from your Start menu. Press Enter to accept the defaults until it asks for the password you set during installation.

•**macOS / Linux:** Open your terminal. By default, PostgreSQL creates a user account named postgres. Switch to that user and launch psql:

```
sudo -i -u postgres
Psql
```

(Note: If you used Homebrew or Postgres.app on Mac, you can usually just type psql postgres in your normal terminal).

You should now see the prompt change to postgres=#, indicating you are connected as the superuser.

Create a Database and User

It is best practice not to use the default postgres superuser for everyday tasks. Let's create a new database and a new user.

Type the following commands into your psql prompt (don't forget the semicolons at the end of SQL statements!):

1. **Create a new database:**

```
SQL
```

```
CREATE DATABASE my_first_db;
```

2. **Create a new user with a password:**

```
SQL
```

```
CREATE USER my_user WITH ENCRYPTED PASSWORD 'my_password';
```

3. **Grant privileges to the user for the database:**

```
SQL
```

```
GRANT ALL PRIVILEGES ON DATABASE my_first_db TO my_user;
```

To connect to your newly created database, use the `\c` (connect) meta-command

```
SQL
```

```
\c my_first_db
```

Your prompt should change to `my_first_db=#`.

Create a Table:

```
CREATE TABLE employees (
    id SERIAL PRIMARY KEY,
    first_name VARCHAR(50),
    last_name VARCHAR(50),
    email VARCHAR(100) UNIQUE,
    hire_date DATE
);
```

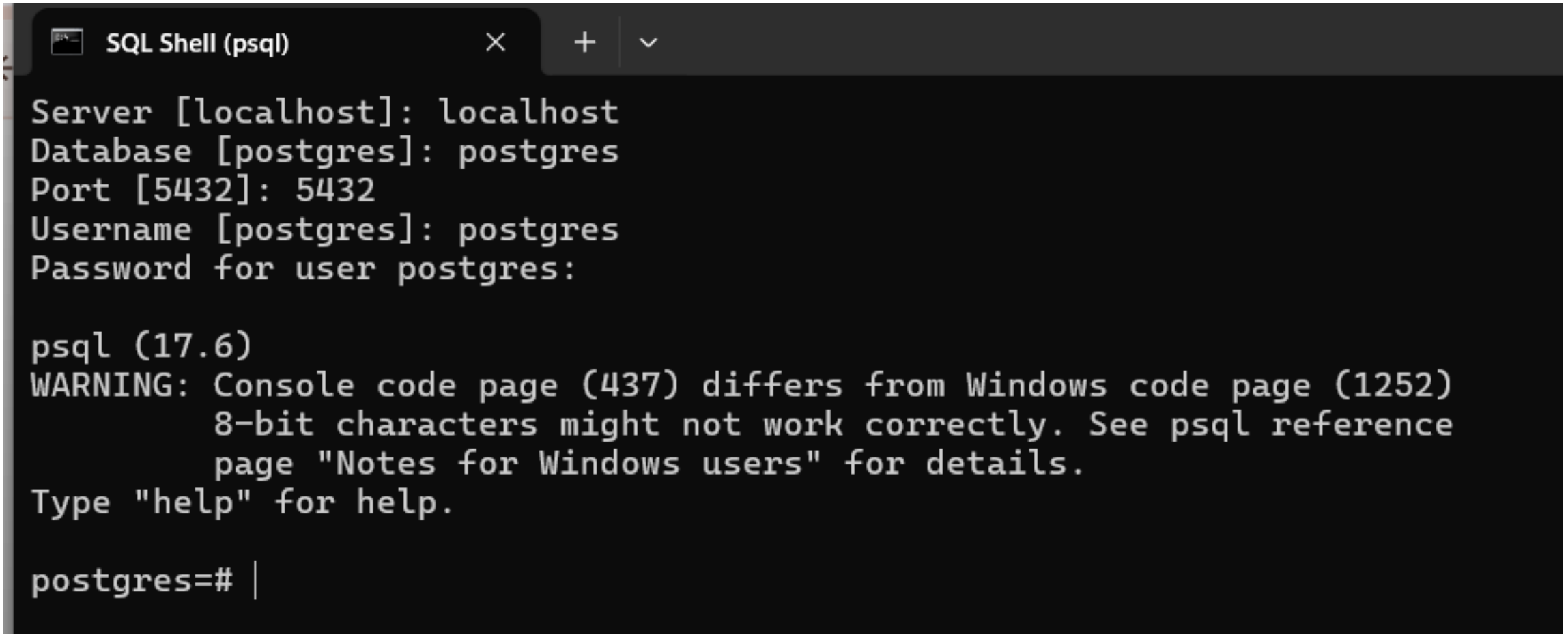
Task 1: Using pgAdmin4, how to create our own table?

- Databases
  - postgres
    - Schemas
      - public
        - tables(right click)
          - Query tool (Query editor displayed)

SQL Command to create your own table:

* create table student (ID integer, name varchar(20));

Task 2: How to Connect with PostgreSQL server Using SQL Shell(Psql)?



Task 3: List : list of available databases under the postgres server using list command.

- Command : \l.

```
postgres=# \l
                                                    List of databases
   Name    |  Owner   | Encoding | Locale Provider |          Collate           |           Ctype            | Locale |
ICU Rules |   Access privileges
-----------+----------+----------+-----------------+----------------------------+----------------------------+--------+-
----------+-----------------------
 DVDRental | postgres | UTF8     | libc            | English_United States.1252 | English_United States.1252 |        |
           |
 postgres  | postgres | UTF8     | libc            | English_United States.1252 | English_United States.1252 |        |
           |
 template0 | postgres | UTF8     | libc            | English_United States.1252 | English_United States.1252 |        |
           | =c/postgres          +
           |          |          |                 |                            |                            |        |
           | postgres=CTc/postgres
 template1 | postgres | UTF8     | libc            | English_United States.1252 | English_United States.1252 |        |
           | =c/postgres          +
           |          |          |                 |                            |                            |        |
           | postgres=CTc/postgres
 testdb    | postgres | UTF8     | libc            | English_United States.1252 | English_United States.1252 |        |
           |
(5 rows)


postgres=#
```

Task 4: See list of tables under the postgres database

- Command : \dt

```
postgres=# \dt
           List of relations
 Schema |   Name    | Type  |  Owner
--------+-----------+-------+----------
 public | student | table | postgres
(1 row)


postgres=#
```

Task 5: How to create database from **SQL Shell**

Step 1: Create your own database using the following command.

- Command : create database < database name>;

  Example: create database **TestDB**;

Step 2: Since you login to access the default database (postgres), we have to change the search path to our own database using the following command.

Command : set search_path to < your databse name>;

set search_path to TestDB

Step 3: check the connectivity of our own databse to the postgres server, use the following command.

- Command: **select version();**

Screenshot of the command: select version();

```
postgres=# create database TestDB;
CREATE DATABASE
postgres=# set search_path to TestDB;
SET
postgres=# select version();
                                                  version
-------------------------------------------------------------------------------------------------
 PostgreSQL 17.6 on x86_64-windows, compiled by msvc-19.44.35213, 64-bit
(1 row)
```

Step 4: To see list of default and our databases under the server using list command:

Command: \l

```
postgres=# \l
                                                                 List of databases
   Name    |  Owner   | Encoding | Locale Provider |          Collate           |           Ctype            | Locale
ICU Rules |   Access privileges
-----------+----------+----------+-----------------+----------------------------+----------------------------+--------
----------+-----------------------
 DVDRental | postgres | UTF8     | libc            | English_United States.1252 | English_United States.1252 |
           |
 postgres  | postgres | UTF8     | libc            | English_United States.1252 | English_United States.1252 |
           |
 template0 | postgres | UTF8     | libc            | English_United States.1252 | English_United States.1252 |
           | =c/postgres          +
           |          |          |                 |                            |                            |
           | postgres=CTc/postgres
 template1 | postgres | UTF8     | libc            | English_United States.1252 | English_United States.1252 |
           | =c/postgres          +
           |          |          |                 |                            |                            |
           | postgres=CTc/postgres
 testdb    | postgres | UTF8     | libc            | English_United States.1252 | English_United States.1252 |
           |
(5 rows)


postgres=#
```

Task 6: load **Dvdrental** database using **SQL Shell**

Step 1: create your database using the following command:

Command: create database **DvdRentalTwo;**

Step 2: Check it by using the list command:

Commnad: \l

See next page.

```
postgres=# create database DvdRentalTwo;
CREATE DATABASE
postgres=# \l
                                                                    List of databases
     Name     |  Owner   | Encoding | Locale Provider |         Collate           |           Ctype           | Locale
 | ICU Rules |   Access privileges
--------------+----------+----------+-----------------+---------------------------+---------------------------+-------
-+-----------+-----------------------
 DVDRental    | postgres | UTF8     | libc            | English_United States.1252 | English_United States.1252 |
 |           |
 dvdrentaltwo | postgres | UTF8     | libc            | English_United States.1252 | English_United States.1252 |
 |           |
 postgres     | postgres | UTF8     | libc            | English_United States.1252 | English_United States.1252 |
 |           |
 template0    | postgres | UTF8     | libc            | English_United States.1252 | English_United States.1252 |
 |           | =c/postgres          +
              |          |          |                 |                           |                           |
 |           | postgres=CTc/postgres
 template1    | postgres | UTF8     | libc            | English_United States.1252 | English_United States.1252 |
 |           | =c/postgres          +
              |          |          |                 |                           |                           |
 |           | postgres=CTc/postgres
 testdb       | postgres | UTF8     | libc            | English_United States.1252 | English_United States.1252 |
 |           |
(6 rows)


postgres=#
```

Task 6: load **Dvdrental** database using **SQL Shell**

Step 3: From SQL Shell go to CMD to Rest the search path to our own databse :

Command:

C:\Program Files\PostgreSQL\17\bin pg_restore –h localhost –d dvdrentaltwo –U postgres –p 5432

C:\Users\Wondim\Downloads\dvdrental \dvdrental.tar

```
C:\Program Files\PostgreSQL\17\bin>pg_restore -h localhost -d dvdrentaltwo -U postgres -p 5432 C:\Users\Wondim\Downloads
\dvdrental\dvdrental.tar
Password:


C:\Program Files\PostgreSQL\17\bin>
```

Task 6: load **Dvdrental** database using **SQL Shell**

Step 4: From CMD , Go to SQL Shell to check the connectivity of our new databse to the server

Command: \c dvdrentaltwo

```
postgres=# \c dvdrentaltwo
You are now connected to database "dvdrentaltwo" as user "postgres".
dvdrentaltwo=#
```

Task 6: load **Dvdrental** database using **SQL Shell**

- **Step 5: Check the list of tables under our new database**

```
dvdrentaltwo=# \dt
             List of relations
 Schema |     Name      | Type  |  Owner
--------+---------------+-------+----------
 public | actor         | table | postgres
 public | address       | table | postgres
 public | category      | table | postgres
 public | city          | table | postgres
 public | country       | table | postgres
 public | customer      | table | postgres
 public | film          | table | postgres
 public | film_actor    | table | postgres
 public | film_category | table | postgres
 public | inventory     | table | postgres
 public | language      | table | postgres
 public | payment       | table | postgres
 public | rental        | table | postgres
 public | staff         | table | postgres
 public | store         | table | postgres
(15 rows)


dvdrentaltwo=#
```